The Persistence Boundary: A Measurable Spectral Law for the Whitening Knee in Muon-style Optimization

Working draft v3 — 2026-08-25

Status: working draft v3 (2026-08-25). All numbers measured in this repository; probe IDs refer to `research_runs/probe_*` logs and `research/OPTIMIZER_RESEARCH.md` §0.5. Figures render from raw logs via `experiments/paper_figs.py` (no manual plotting).

> **v2 → v3: the tracked gaps are closed.** 1. The U-curve is now single-frame: classic / cubic5 / b10 were re-run post-fix (P31–P33); the five-point U lives entirely in the current frame (§5.2). 2. Both arms and the peak now have 3 seeds each (P25, P36–P39); every arm-vs-peak difference is positive on every seed (§5.2, §6). 3. Dense platform: the knee grid is now 5 points (b002/b005/b01/b05/b10; P34/P35 added) **plus a 2000-step paired tier (P40–P42) that reverses the 500-step dense ranking** — the 500-step winner b002 destabilizes (spike at step ≈600, NaN by ≈1000, paired control clean on identical data) and b005 concedes +0.072 to b05. Warmup knee optima bias downward; the steady-state optimum sits at the band's upper edge on both architectures (§5.4). The MoE-side 2000-step knee check (P43 b10 vs. P28a b05) shows the mirror asymmetry: b10's +0.08 warmup penalty closes to -0.02 by step 1900 — under-whitening fades, injection compounds; the b05 default now rests on 2000-step evidence. 4. The transition-band upper edge has an operational definition — first log-σ bin whose excess persistence (median ρ - median floor) reaches ≥ 0.9 — and all σ^*_{upper} values are recomputed under it (`experiments/sigma_upper*.py`; §4.3). 5. §7's per-head row is "consistent, not confirmed" (softened; the pre-fix harm did not reproduce post-fix).
>
> **Correction in v3.** The §4.1 KS statistic against an MP bulk for the 768×768 group was numerically invalid (uniform-grid CDF integration diverges under the c = 1 MP density singularity at y = 0; on synthetic pure-MP data the same code reports D ≈ 0.60). The rejection of MP calibration rests on the remaining, unaffected evidence (§4.1).

Abstract

Muon-style optimizers orthogonalize the momentum matrix with Newton–Schulz (NS) iterations whose scalar spectral response is a design choice. Recent relaxed schedules leave a *knee*: singular values below a threshold receive near-zero weight instead of being flattened to 1 — and are both cheaper and better. But where should the knee sit? We turn this hyperparameter question into a measurement question. First, we show the natural closed-form answer from random-matrix theory (knee = Marchenko–Pastur bulk edge λ_+) fails on real training momentum: spectra are smooth heavy-tailed (Zipf slopes -0.7…-2.7), entry noise is anisotropic at ratios of 10^2–10^4, and spike peeling diverges. Second, we show the question can still be answered assumption-free: cross-microbatch direction persistence $\rho(\sigma)$ collapses sharply at a measurable boundary σ^*, stationary across our training window (micro-steps 300→1900). Third, we state and validate the resulting law: σ^* is the *lower* edge of a transition band whose persistent-signal fraction climbs from ≈0.3 to ≈1; the quality-optimal knee sits at the band's *upper* edge, and the two arms of the U-shaped quality curve are exactly the band's width (+0.05 nat when the knee is at the lower edge; +0.09 when past the upper edge). The law replicates across architectures — with a horizon twist. On a 132M dense model σ^* is ~10× lower than on a 1080M MoE, and the warmup-tier optimum moves with it (five-point grid); but by 2000 steps the dense ranking

reverses — the lowest knee destabilizes (NaN after a step-600 spike), the mid-band point concedes 0.07 nat, and the shared steady-state optimum sits at the band's upper edge (knee50 $\approx$ 0.01) on *both* architectures. The asymmetry is structural: under-whitening is paid upfront, variance injection compounds. A method built on the naive binary rule (per-shape knees at σ^*, "EdgeCubic") is neutral at 500 steps but degrades monotonically by 2000 steps (+0.47 nat on the auxiliary MTP loss), with the damage localized in exactly the group the corrected law flags — a controlled negative result that doubles as the law's stress test. Six previously reported failure modes of spectral shaping, plus Pion's RLVR observation, admit a single zero-parameter account. Practically: the knee is not a constant to tune but a boundary to measure — and in our stack the measured optimum coincides with the existing cubic5b05 schedule, which we retain as default.

1. Introduction

The Muon family replaces the update direction for a 2D weight matrix by an orthogonalization of its momentum $\mathrm{M} = \mathrm{U}\Sigma\mathrm{V}^\mathsf{T}$. The polar factor $\mathrm{U}\mathrm{V}^\mathsf{T}$ is never computed exactly: a fixed Newton–Schulz polynomial iteration reshapes the singular values through a scalar response $\tau(\sigma)$. "How Much Orthogonalization Does Muon Need?" (arXiv 2606.00371) showed that training quality is *not* monotone in the accuracy of this map: a relaxed-cubic schedule with a pronounced knee (small σ lifted only to $\approx$0.12 instead of $\approx$1) is both cheaper and better. Pion (arXiv 2605.19282) independently found that suppressing the small-σ tail helps in low-SNR regimes (RLVR, VLA). Neither work says where the knee *should* sit; both inherit it from numerical considerations (bf16 resolution).

This paper answers "where is the knee" with a law that is:

- **Measurable.** The boundary σ^* is read directly off cross-microbatch gradient correlations, with no distributional assumption (§2.3, §4).
- **Stationary.** σ^* does not drift over the probed training window (micro-steps 300–1900; §4.3).
- **Intervention-validated.** Quality vs. knee position is U-shaped, and the U's arms coincide with the measured *transition band* $[\sigma^*,\ \sigma^*_{\text{upper}}]$: knees at the lower edge over-inject non-persistent directions, knees past the upper edge under-whiten persistent ones (§5).
- **Cross-architecture.** On a 132M dense model σ^* is $\approx$10× lower than on a 1080M MoE, and the empirically optimal knee moves by the same factor — the law replicates as "optimum = measured band", not as a universal constant (§5.4).
- **Unifying.** Six in-house failure/quality results and one literature observation (Pion on RLVR) follow from one mechanism with zero fitted parameters (§7).

We also report the theory that motivated the measurement — an equivariance classification of update rules and a closed-form shrink-then-whiten optimum $t^*(\sigma)$ under a spiked model (§3) — and *where it breaks*: real momentum is neither spiked nor isotropic, so the closed form calibrates incorrectly while its shape intuition survives (§4.1). A deliberate method attempt built on the naive reading of the law ("EdgeCubic", per-shape knees at σ^*) fails at longer horizons in exactly the way the corrected law predicts (§8). We count this as evidence: the law is falsifiable in both directions, and it called the failure's location in advance.

Contributions.

1. A snapshot-and-persistence measurement protocol (momentum + micro-batch gradients $\to \rho(\sigma) \to$ per-shape-group σ^*), with the analysis chain validated on synthetic spiked-MP ground truth ($\hat{\lambda}_+$ within 2%, spike ρ 0.91 vs. BGN-theoretical 0.87–0.96).
2. The persistence-boundary law for the whitening knee — including the transition-band refinement and its U-curve signature — validated by intervention on two architectures.
3. A zero-parameter unified account of six failure modes plus one literature observation.
4. A controlled negative result (EdgeCubic at 2000 steps) with single-variable attribution, doubling as the law's stress test.
5. Practical guidance: what to measure before touching the knee; why 500-step promotion gates are unsafe; why cubic5b05 is already near-optimal in our regime.

2. Background and setup

2.1 Spectral shaping in Muon-style optimizers

For a 2D weight matrix with momentum M (EMA + nesterov), Muon steps along Orth(M) ≈ NS(M). Every published variant is a spectral map — singular vectors kept, spectrum reshaped by a scalar response:

variant	response	GEMMs/matrix	knee50
classic NS5 (quintic, 5 steps)	band [0.65, 1.20], τ(1e-3)=0.47	15	≈0.001
Polar Express (PE5)	band [0.80, 1.13], τ(1e-3)=0.86	15	≲0.001
cubic5 (relaxed, arXiv 2606.00371)	band [0.7, 1.3], τ(1e-3)=0.12	10	≈0.004
cubic5b05 (our default)	same generator, l0=0.05	10	≈0.010
cubic5b10 / b002	l0=0.1 / l0=0.002 (7 steps)	10 / 14	≈0.02 / ≈5e-4

knee50 := the F-normalized singular value receiving half weight, measured numerically from each schedule's scalar response (it tracks the design lower-clamp l0 sub-linearly and is the variable we intervene on). Our optimizer ("MuonH") additionally projects each updated matrix back to its pre-update Frobenius norm (hyperball pinning); this detail interacts with one bug the measurement uncovered (§4.4).

2.2 Experimental platforms and noise discipline

- **MoE probe platform**: 1080M params — hidden 768, 8 layers + 1 MTP block (depth 1), 8 heads, 256 routed experts top-8 + 2 shared (intermediate 384), KDA linear-attention layers interleaved; sequence 1024 with document packing, micro-batch 12 × accumulation 2 (24,576 tokens per optimizer step); formula-derived lr (adam 1.545e-3, muon 6.695e-3), warmup 279 micro-steps; bf16, mx.compile, Apple M4 Max.
- **Dense platform** (externalization): 132M params — hidden 768, 12 layers, 12 heads, dense FFN 3072, no MTP/latent path.
- **Tiers**: screening = 500 micro-steps (≈14 min); confirmation = 2000 micro-steps (≈45 min). Every quality claim is a same-seed (1337), same-data-order **paired** comparison. Paired noise floor: ±0.05 nat (Metal nondeterminism only). Cross-seed floor: $\sigma = 0.11$ nat (3 seeds: 5.264 / 5.094 / 5.302, P25). Each claim below states which floor applies.
- **Snapshot tooling** (`trainer/snapshot.py`; env-gated, zero overhead when off): at chosen optimizer steps, dump the momentum fed into NS and the 2×accum independent micro-batch gradients forming consecutive updates, with structured subsampling for stacked expert tensors. Analysis: `experiments/mp_fit.py`.

2.3 The measurement: direction persistence $\rho(\sigma)$

Split the gradients of two consecutive optimizer updates into their constituent micro-batches (after accumulation rescaling). For each F-normalized momentum matrix, take its singular directions $\{u_i v_i^\intercal\}$ at a snapshot step and measure how much of the *subsequent* micro-batch gradient energy aligns with direction i, in excess of a chance-level floor estimated from the same micro-batches; bin by σ. Persistent directions — signal, by definition: the gradient keeps returning to them — have $\rho \approx 1$; fresh-noise directions have $\rho \approx 0$. The boundary σ^* is the collapse point of $\rho(\sigma)$.

The estimator is validated end-to-end on synthetic spiked-MP ground truth: bulk $\rho \approx 0$, spike $\rho = 0.91$ against the BGN-theoretical 0.87–0.96, and the recovered bulk edge $\hat{\lambda}_+$ within 2% of truth. Nothing in the estimator assumes MP, spikes, or isotropy — which §4 shows is essential.

3. Theory: the search space, the null model, and the closed-form optimum

This section compresses `research/SPECTRAL_THEORY.md` §1–§3, keeping what the experiments later confirm or refute. Readers who want only the empirical law can jump to §4.

3.1 Classification: equivariant stateless rules are joint spectral maps

Let Φ: $\mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$ act on momentum. If Φ is **equivariant** (Φ(QMR) = Q·Φ(M)·R for all orthogonal Q, R) and **stateless** (depends only on the current M), then by von-Neumann-type representation theorems Φ is a joint spectral map: Φ(M) = U·diag(τ)·V^T with τ a function of the spectrum. Every published variant (classic NS5, Polar Express, the cubic family, Pion, fractional-power iterations) lives in this space; the literature restricts itself to *separable polynomial* maps τ_i = p(σ_i), a proper subset — joint, spectrum-adaptive maps are equally legal. Leaving the space requires one of three channels: breaking equivariance (row statistics — NorMuon, RowFloor, Aurora, per-head), adding state (second moments, SOAP bases, error feedback), or adding curvature (Mousse). This paper stays inside and asks: what is the optimal τ?

3.2 Null model: momentum = persistent signal + fresh isotropic noise

Model the F-normalized momentum as X = S + E: S low-rank and persistent, E i.i.d. entry noise with variance w/(mn) (w = noise fraction of Frobenius energy). EMA with decay β averages noise down by the effective sample size ESS = (1+β)/(1-β) (39 at β=0.95) while signal stacks coherently. Random-matrix theory (Marchenko–Pastur; BBP; Benaych-Georges–Nadakuditi) then gives: a noise bulk with upper edge $\lambda_+ = \sqrt{w}\cdot\mathbf{(1/\sqrt{m} + 1/\sqrt{n})}$; directions below the edge asymptotically orthogonal to signal; closed-form cosine overlaps above it. A toy EMA model confirms the machinery end to end: predicted vs. measured noise level ratio 0.997; overlap collapse at the predicted edge (0.757 measured vs. 0.760 theory at σ/edge = 1.27; 0.10 at σ/edge = 0.98).

3.3 The in-space optimum: shrink-then-whiten

The Muon ideal is the polar factor of the *signal* — whiten all true descent directions equally. Acting only along observed directions, both least-squares projection onto the signal polar factor and matched filtering give the same shape: $\tau_i = \cos\theta_u(\sigma_i)\cdot\cos\theta_v(\sigma_i)$. Hence the closed form $t^*(\sigma)$: zero below λ_+; partial weight just above it (a direction that is mostly noise gets exactly its overlap fraction); →1 for $\sigma \gg \lambda_+$.

Three properties. (i) **Shape**: a knee plus a flat band, with knee position and transition profile fixed by (m, n, w) — no hyperparameter. (ii) **Legitimacy**: t^* is monotone and bounded, hence realizable as the regularized linear-maximization solution of a convex spectral penalty (a hard knee + flat band is exactly "spectral-norm ball ∩ nuclear-norm penalty"), so a mirror-descent geometry exists and standard convergence theory applies — it is not a heuristic. (iii) **Relation to Gavish–Donoho**: GD shrinkage reconstructs the signal matrix ($\tau_{\mathrm{GD}} = \hat{s}\cdot\cos\theta_u\cdot\cos\theta_v$); our target is the *whitened* signal (drop $\hat{s}$). To our knowledge this "optimal shrinkage under a whitening objective" is absent from both the RMT and the Muon literature.

Compatibility with the impossibility theorem. Muon-p (arXiv 2606.13867) rules out fixed univariate polynomial iterations converging to fractional powers σ^p. t^* is unaffected: it tends to a constant at large σ — precisely the natural attractor of NS-type iterations. In this reading the cubic family is a coarse composite realization of a knee map whose knee is placed by numerics rather than by statistics.

3.4 Where the mathematics allows cheapness

The cost breakpoint is knee sharpness (Bernstein-ellipse argument). A single-shot polynomial approximating a knee at ℓ needs degree ~$1/\sqrt{\text{smoothing radius}}$: measured, a smoothed t^* with knee at 0.029 still has 0.33 error and 0.32 noise leakage at deg-8 (10 GEMMs). Iterated composition grows effective degree exponentially (k cubic steps ≡ degree 3^k), cost ~log(1/ℓ). The cubic iteration machinery is therefore the right substrate; the theory re-prices only its l0 constant — and higher knees need *fewer* steps (l0 = 0.03–0.09 ⇒ 3–4 steps, 6–8 GEMMs vs. 10). Cost and quality point the same way, *if* the knee is placed correctly. §8 tests exactly this.

4. Measurement: real momentum and the persistence boundary

4.1 The closed form fails on real momentum — and why

Running the snapshot pipeline on the MoE platform (probe_p18_snap, optimizer steps 150/300; both time points agree) kills the spiked-MP calibration on all three of its assumptions:

- **Spike peeling diverges** ($\hat{w} \to 0$) for almost every shape group. (An earlier draft cited a KS D = 0.60 against an MP bulk for the one nominally fittable group, 768×768; that statistic was numerically invalid — uniform-grid CDF integration diverges under the c = 1 MP density singularity at y = 0, reporting D ≈ 0.60 even on synthetic pure-MP data. The peeling divergence stands on its own.)
- **Spectra are smooth heavy-tailed** — Zipf slopes -0.7…-2.7 in log σ–log rank, no plateau bulk, no low-rank spike structure.
- **Entry noise is strongly anisotropic** — centered-residual row/column variance ratios of 10^2–10^4 (rows up to 3.2e4, columns up to 2.6e4), violating the i.i.d. premise behind $\lambda_+ = \sqrt{w}(1/\sqrt{m} + 1/\sqrt{n})$.

Both failure branches flagged in advance in the theory note (anisotropy; non-spiked signal) are the ones reality takes. The closed form therefore cannot *place* the knee. What survives is t^*'s shape logic: weight each direction by how much of it is signal — which is exactly what $\rho(\sigma)$ measures, without a model.

4.2 $\rho(\sigma)$ collapses sharply: the boundary σ^* exists

On real momentum the persistence measure collapses cleanly (F-normalized spectrum; snapshot steps 150 and 300 agree):

shape group	measured σ^*
attention q/k/v 768×768	0.001–0.003
o_proj 768×384 / 384×768	≈0.003
KDA gates 96×768 / 768×96	0.001–0.003
routed experts gu 768×384	0.003–0.01 (softer collapse)
routed experts dw 384×384	0.005–0.01
shared FFN 768×1536 / 1536×768	3e-4–1e-3 (lowest)
MTP block 768×2304 / 2304×768	3e-4–1e-3 (lowest)
kv_up 128×768	no collapse in view ($\rho > 1$ down to $\sigma_{\min} = 0.024$)

σ^* is *not* monotone in matrix size or aspect ratio: the within-MoE variation (30×) is as large as the cross-architecture shift (§5.4). σ^* must be measured per shape group; it is neither a universal constant nor an MP formula.

4.3 σ^* is stationary, and the collapse is a band, not a step (P30)

A natural worry: σ^* measured during warmup may expire later. Late-run snapshots (optimizer steps 500/750/950 = micro-steps 1000/1500/1900; probe_p30_latesnap) show the boundary does not move — shown here for the two lowest-σ^* groups, the ones §8 turns on:

group	σ bin	signal fraction @500	@750	@950
MTP 768×2304	[1e-5, 3e-4)	0.15	0.07	0.06
	[3e-4, 1e-3)	0.37	0.31	0.36
	[1e-3, 3e-3)	1.07	0.92	1.00
	[3e-3, 1e-2)	1.34	1.17	1.29

group	σ bin	signal fraction @500	@750	@950
shared FFN 768×1536	[3e-4, 1e-3)	0.76	1.13	0.78
	[1e-3, 3e-3)	1.25	1.08	1.43

(Fractions are ρ minus the estimator floor; ≈1 = fully persistent.) Two facts in this table drive everything downstream: **(i)** the collapse is sharp in log σ, and stationary across micro-steps 300→1900; **(ii)** it is *not* a step — the signal fraction climbs gradually (≈0.3 → 1.0 → 1.3) across nearly a decade for the heavy-tailed groups. We call [σ^*, σ^*_{upper}] the **transition band**.

Operational band edges (v3; replaces eyeballing). Bin per-direction values by log σ (bins [1e-5, 3e-4, 1e-3, 3e-3, 1e-2, …], ×√10); per bin take the *median* ρ and *median* floor (the median is essential — the floor blows up at tiny σ); excess := median ρ - median floor. σ^*_{upper} := the lower edge of the first bin with excess ≥ 0.9; σ^* := the lower edge of the first bin with excess ≥ 0.3. Recomputed over all groups and snapshot steps (`experiments/sigma_upper.py` on the p18 dumps, `experiments/sigma_upper_from_txt.py` on the p30 tables):

shape group	σ^*	σ^*_{upper} (steps 500/750/950)
MTP 768×2304 / 2304×768	3e-4–1e-3	1e-3 / 1e-3 / 1e-3 — stationary
shared FFN 768×1536 / 1536×768	3e-4–1e-3	1e-3 / 3e-4 / 1e-3
attention q/k/v 768×768	0.001–0.003	no crossing below 1e-2 (excess 0.70–0.83 in [1e-2, 3e-2))
o_proj 768×384 / 384×768	≈0.003	≈3e-3, noisy at later steps
KDA gates 96×768 / 768×96	0.001–0.003	no crossing ≤ 3e-2 (excess ≲ 0.5)
routed experts gu/dw	0.003–0.01	no crossing ≤ 1e-1 (excess ≲ 0.5)
kv_up 128×768	no collapse in view	—

So the band is narrow for the heavy-tailed low-σ^* groups (upper edge ≈1–3×σ^*) and wide for attention and the expert stacks (upper edge 10×σ^*; for the routed experts the fraction never reaches 0.9 in view — the "band" there spans the measured spectrum, consistent with their softer collapse, §4.2). Where a single knee must serve all groups, the binding constraint is the parameter-dominant attention block, whose upper edge is ≈1e-2.

4.4 The measurement doubles as a diagnostic: a silent training bug

The step-150/300 snapshots showed 20/20 sampled gate_down matrices with identically zero gradient — impossible unless something upstream is pinned. Cause: hyperball pinning projects every update back to the pre-update Frobenius norm, so a zero-initialized matrix is pinned at radius 0 *forever*. GatedNorm.gate_up (zero-init by design) had therefore been frozen in every historical run, the gate stayed at identity, and gate_down received no gradient. Fix: gate_up moved to the AdamW group (standard "zero-init gates go to Adam" practice) plus a zero-norm guard in the projection; regression test `test_muonh_zeroinit.py`. All probe numbers in this paper are post-fix unless explicitly labeled; the fix shifted the baseline by ≈0.04 nat, so pre-fix and post-fix numbers are never subtracted across frames.

5. The law

5.1 Statement

Definitions. For each shape group: σ^* := the collapse point of $\rho(\sigma)$ (§2.3); transition band := [σ^*, σ^*_{upper}] under the operational thresholds of §4.3 (first bin with excess ≥ 0.3 / ≥ 0.9); knee50 := the σ receiving half weight under a given spectral response.

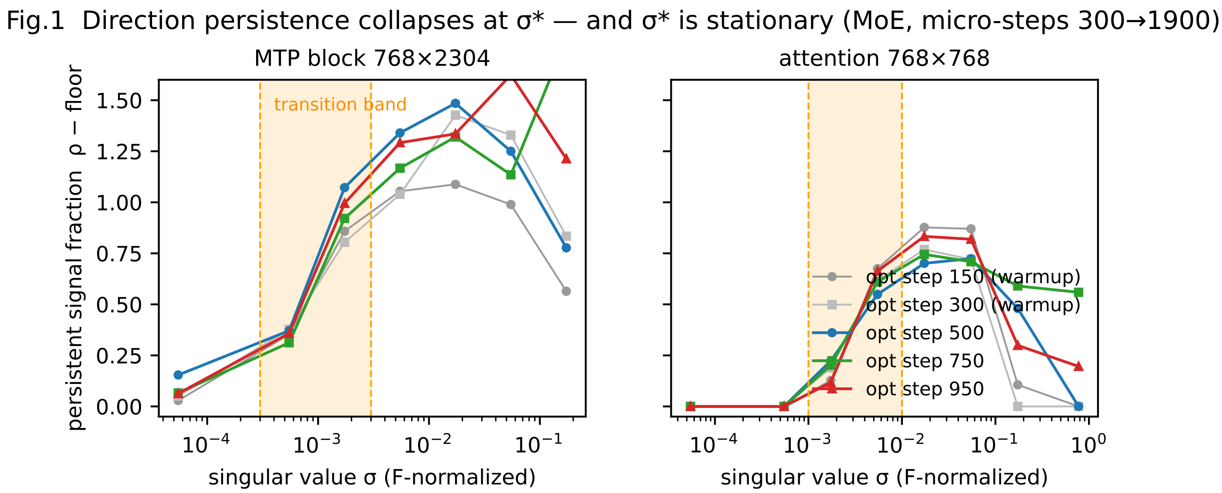

Figure 1: Fig.1: $\rho(\sigma)$ collapse and stationarity

Persistence-boundary law (corrected form). Give zero weight below σ^*; inside the transition band, shrink by the measured signal fraction; flatten above σ^*_{upper}. A single-knee schedule should place knee50 at the **upper** edge σ^*_{upper} — not at σ^*. Deviations are penalized on both sides, and the two arms of the resulting U-shaped quality curve span exactly the band: knees at σ^* hand 30–70%-noise directions full weight (variance injection); knees past σ^*_{upper} under-whiten fully persistent directions.

The naive reading (binary: zero below σ^*, flat above) is what §8 falsifies; the corrected form is what the intervention data below supports.

5.2 Intervention: the U-curve on the MoE (500-step tier)

Only the knee of the cubic schedule is varied. Since v3 the full U lives in a single frame (post-fix; the pre-fix frame is retained only for the qualitative extremal points marked †). Paired, same seed (1337), baseline b05 = 5.264, paired noise floor ± 0.05:

probe	knee50	loss@500	Δ vs b05	reading
classic NS5 (P31)	≈0.001	5.358	**+0.094**	far left arm
cubic5b002 (7 steps, P20)	≈5e-4 (at σ^*)	5.318	**+0.054**	left arm: transition band at full weight
cubic5 (P32)	≈0.004	5.274	+0.010	left shoulder
cubic5b05 (default, P19)	≈0.010 (≈σ^*_{upper})	**5.264**	—	**peak**
cubic5b10 (P33)	≈0.02	5.349	**+0.085**	right arm
PE5 (sharp knee) †pre-fix	≲0.001	+0.14 vs. classic		far left; sharper is worse
no-NS (norm only) †pre-fix	none	+0.34 vs. classic		no whitening at all
frac family (no flat top) †pre-fix	—	+0.20…0.28		respects the knee but never flattens — right-arm failure by construction

Seeds. The peak and both arms now have 3 seeds each (P25, P36–P39). Per-seed Δ vs. b05 — b002: +0.054

/ +0.014 / +0.024 (seeds 1337/1338/1339); b10: +0.085 / +0.065 / +0.038. The arm penalties are positive on **every** seed; cross-seed σ of the Δ is ≈0.02 nat for both arms. The monotone trio τ(1e-3) = 0.86 / 0.47 / 0.12 → +0.14 / 0 / -0.08 (PE5 / classic / cubic5) was the original smoking gun.

Arms = band width. Left arm: a knee at $\sigma^* \approx$ 5e-4 hands the band [σ^*, ≈3e-3] — measured 30–70% non-persistent for the heavy-tailed groups — full weight. Right arm: a knee at $0.02 > \sigma^*_{\text{upper}}$ under-whitens the range [3e-3, 1e-2] whose measured signal fraction is ≈1. The peak sits between, at the upper edge of the parameter-dominant attention band (§4.3). This is the measurable version of t^*'s shape intuition.

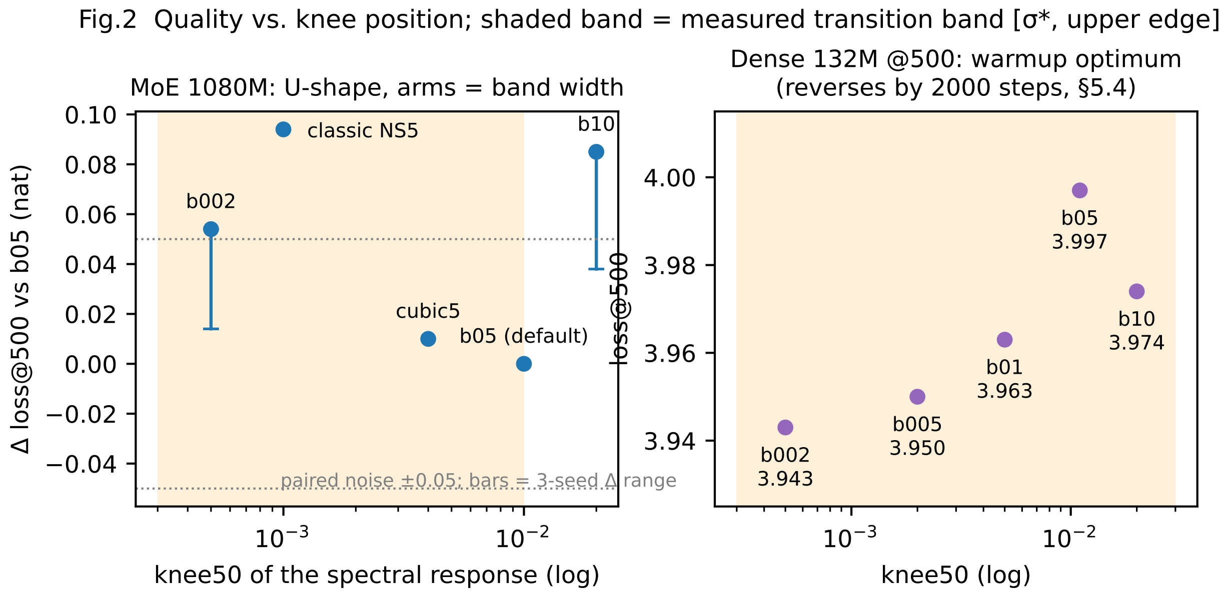

Figure 2: Fig.2: U-curve on two architectures

5.3 Prediction scorecard (pre-registered in the theory note)

#	prediction	verdict
1	optimal knee ≈ λ_+ closed form; raising the knee helps up to λ_+	**half**: raising helps up to b05 then U-turns (b10 +0.09) ✓; the closed-form calibration fails (§4.1) × — replaced by measured σ^*
2	direction correlation collapses at $\hat{\lambda}_+$	**form right, position corrected**: $\rho(\sigma)$ collapses (sharper than predicted), but at $\sigma^* \neq \hat{\lambda}_+$
3	per-shape knee scales as $(1/\sqrt{m} + 1/\sqrt{n})$	**not confirmed**: measured σ^* does not follow MP ratios (§4.2); per-head compensation is unnecessary post-fix (§6)
4	knee scales as $\sqrt{B}$ with batch size	**not confirmed** at B/2 (P22): b07 vs. b05 within noise; the lever may be ESS-dominated at fixed β
5	residual headroom below the current knee region	**direction right**: both arms harmful, interior peak (P17/P20) — U confirmed

Zero of two quantitative scaling laws survived; the entire qualitative frame (U-shape, knee = persistence boundary, zero weight below σ^*) did. Hence the paper's form: a measurement-plus-intervention law rather than a closed-form prediction.

5.4 Externalization: the law replicates on a dense model — by moving

Dense 132M platform, same protocol, 500-step tier, seed 1337 (paired floor ±0.05; five-point knee grid, all in the post-fix frame):

schedule	knee50	loss@500	reading
b002 (P26c)	≈5e-4	**3.943**	knee at the dense σ^* — plateau edge
b005 (P35)	≈2e-3	3.950	plateau
b01 (P34)	≈5e-3	3.963	plateau, drifting up
b10 (P26b)	≈0.02	3.974	under-whitening cost visible
b05 (P26a)	≈0.011	3.997	**worst at this tier**

The MoE ranking does **not** transfer at this tier: the MoE optimum b05 is *worst* here, and the best point sits at b002 (vs. b05: -0.054, beyond the paired floor). The three lowest knees (b002/b005/b01) span 0.020 nat — a plateau within the noise floor, i.e. the dense left arm has not yet appeared even with the knee at σ^* itself. Measurement explains the shift before any curve-fitting (probe_p27_dense_snap): dense $\sigma^* \approx$ 3e-4–1e-3 (attention), ≤0.003 (FFN) — an order of magnitude *below* the MoE's 0.003–0.01. What does **not** move is the band's upper region: under the §4.3 rule the dense FFN's excess reaches 0.9 only at ≈3e-2 and the dense attention block never crosses 0.9 in view (0.5–0.87 across [3e-3, 3e-1)) — the dense band is *wide*, overlapping the MoE's 1e-2–3e-2 region. So what the 10× shift moves is the band's **lower** edge.

The 2000-step tier reverses the ranking — and explains it (P40–P42, paired, seed 1337, loss@1900). b05 completes cleanly at **2.905**. b005 completes at 2.977 — **+0.072 worse than b05**, beyond the paired floor. b002, the 500-step winner, first *replicates* its advantage at step 500 of the same run (3.929 vs. b05's 3.994, -0.065, matching the 500-step tier's -0.054), then spikes to 8.36 at step ≈600, half-recovers, and is irreversibly NaN by step ≈1000 (the NaN guard skips non-finite updates but cannot un-poison the parameters). The paired b05 control passes the *identical* data order without any excursion, so the divergence is a property of the schedule, not the data.

Read through §7's mechanism, this is the predicted asymmetry between the two arms' costs. Under-whitening (right arm) is a per-step efficiency loss: paid immediately, absorbable by the learning rate — hence visible already at 500 steps, where b05 pays it. Injecting non-persistent directions (left arm) is variance that *compounds* in the momentum state: nearly free at 500 steps (the plateau reaches all the way down to σ^*), decisive by 2000 (b005 concedes 0.07; b002 destabilizes outright). The warmup tier therefore biases knee selection **downward**, and the steady-state optimum sits higher — on dense at knee50 ≈ 0.01, the same value as the MoE default, inside the wide measured band.

The MoE's right arm shows the same asymmetry in mirror image: b10's warmup penalty (+0.082 at step 500 of the 2000-step run, replicating the 500-step tier's +0.085) fully closes by step 1000 and ends nominally *ahead* at -0.018 at step 1900 — within the paired floor (P43 vs. the P28a baseline 4.087; component split main -0.036 / MTP +0.058). Under-whitening fades; injection compounds. The MoE default b05 is therefore no longer *better* than b10 at 2000 steps, but it is not worse either — the default survives, now on 2000-step evidence rather than warmup evidence.

The §8 lesson — warmup optima are not steady-state optima — thus applies to the headline grid itself, and the law's practical form hardens: **place knee50 at the band's upper edge, never at σ***. (Single seed and one run per point at this tier; the b002 divergence is one event, §10.)

6. Scaling and robustness checks

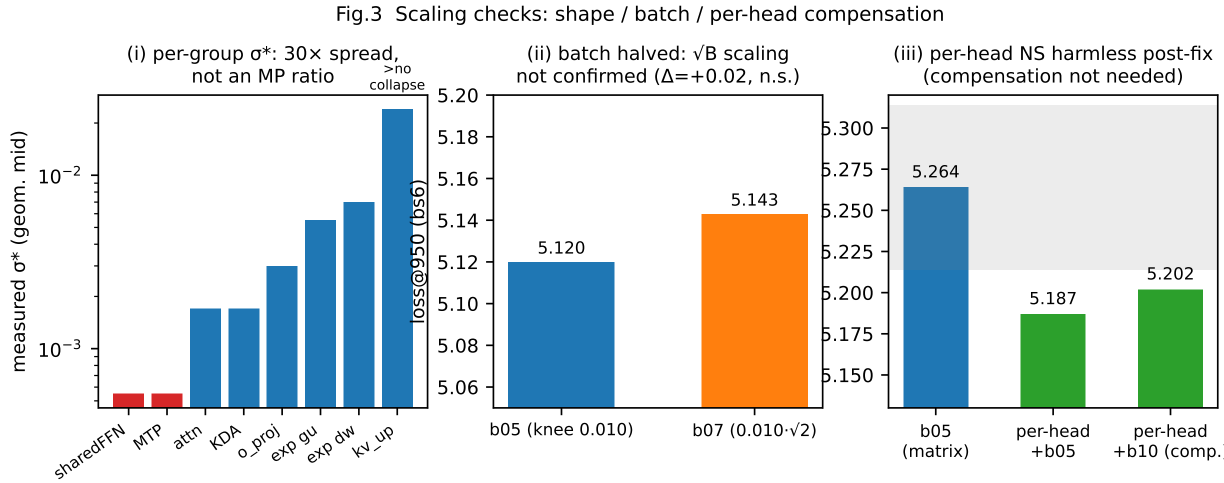

Figure 3: Fig.3: shape / batch / per-head

- **Shape axis** (scorecard row 3). Per-head NS (64×768 blocks instead of 768×768) with MP-motivated knee compensation ×1.9 (P21: 5.202, -0.06) and without (P21b: 5.187, -0.08) are both fine at 500 steps — uncompensated slightly better. The pre-fix stack had per-head at +0.14 harmful; this does not reproduce post-fix and is attributed to interactions in the old (gate-frozen) stack. The strong MP-ratio form of shape scaling is not supported; per-group σ^* measurement is the operative replacement.
- **Batch axis.** Halving the effective batch (bs6 × accum2), b07 (= b05·$\sqrt{2}$) is not better than b05 (+0.02, within noise; P22). The $\sqrt{B}$ prediction is unconfirmed at B/2; at fixed β the noise level may be ESS-dominated rather than batch-dominated.
- **Seeds.** Cross-seed σ of absolute loss = 0.11 nat (P25); cross-seed σ of the *paired* arm-vs-peak Δ is ≈0.02 nat, and both arm penalties are positive on all three seeds (§5.2). All ≤0.09-nat claims at the 500-step tier are *paired* claims (±0.05 floor); we claim the U's direction, not exact nat values, across seeds. The 2000-step effects in §8 (+0.18 total, +0.47 on the MTP component) exceed both floors.

7. A zero-parameter unification of failure modes

One mechanism, no fitted constants: **weight given to zero-overlap (non-persistent) directions is pure variance injection — first-order harmful; bandwidth variation among persistent directions is absorbed by the learning rate — second-order harmless.** The measured σ^* table (§4.2) turns this into per-case arithmetic:

observation	account
PE5 +0.14 (τ(1e-3) = 0.86)	σ = 1e-3 is below every measured attention boundary: a non-persistent direction at 0.86 weight — maximal injection
classic NS5 (τ(1e-3) = 0.47)	the same direction at half weight — between PE5 and cubic5, as observed
cubic5 -0.08 (τ(1e-3) = 0.12)	least injection of the trio; three points monotone in the predicted direction
no-NS, norm-only +0.34	$\tau \propto \sigma$: top directions monopolize the step; mid-strength *persistent* directions underweighted — the opposite failure (under-whitening)

observation	account
stale polar factor +0.4…0.5, worst in warmup	full weight on directions that have *rotated away* — zero-overlap full weight in the extreme; momentum rotates fastest in warmup ⇒ maximal harm there ✓
per-head NS +0.14 (pre-fix stack only)	cutting to 64×768 raises the block's noise edge ≈1.8× (MP estimate), doubling injected noise mass at fixed coefficients. Not reproduced post-fix (§6) — consistent, not confirmed. Also matches Pion's report that per-head helps only combined with a high-pass
Pion's RLVR gains	low-SNR regimes raise the boundary; a fixed classic knee is then *relatively* too low — tail suppression is the predicted fix. Consistent with Pion's gains appearing in RLVR/VLA, not pretraining

Zero-parameter does not mean unfalsifiable: this account predicted the U's left arm before it was measured (P20), predicted the dense-model reversal at the warmup tier (§5.2→§5.4), and predicted the localization of EdgeCubic's failure (§8). Its sharpest implication — injection compounds, under-whitening does not — is what the 2000-step dense re-reversal observes (§5.4).

8. Method attempt and controlled negative result: EdgeCubic

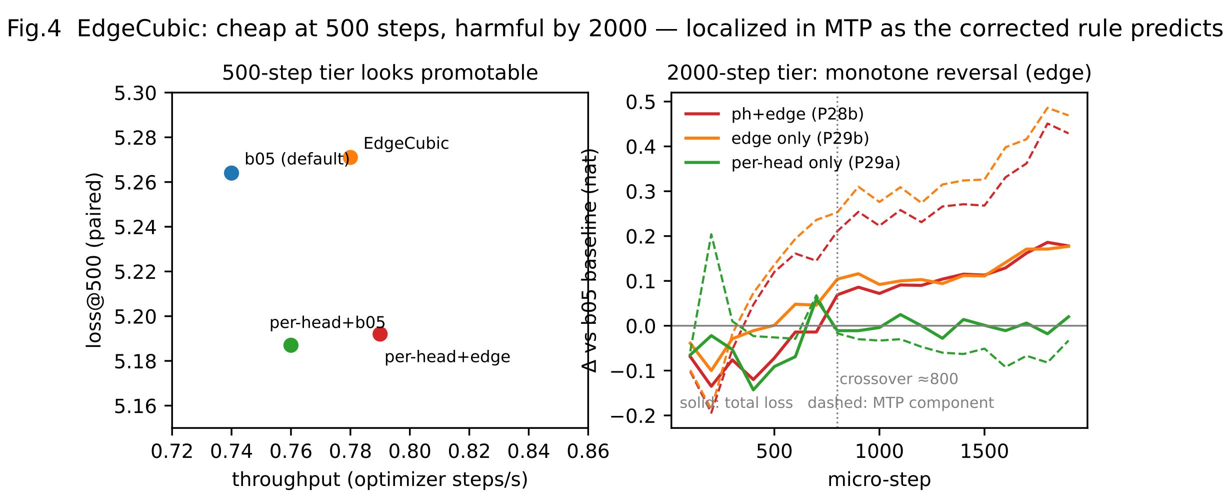

Figure 4: Fig.4: Pareto + 2000-step reversal

The naive engineering reading — set each shape group's knee at its measured σ^* ($l0 = 4.5\sigma^*$, so knee50 $\approx \sigma^*$), let the schedule generator pick its own step count (4–8 steps; fewer GEMMs at higher knees) — is a coefficient-table swap (`VIBY_MUONH_EDGE=1`).

- **500-step tier**: quality-neutral (5.271 vs. 5.264, within the paired floor) at **+5.4% throughput** (0.78 vs. 0.74 step/s; 0.79 with per-head). Passed the pre-registered promotion bar.
- **2000-step tier**: monotone reversal. Δloss crosses zero at step ≈800 and grows to **+0.18 at step 1900**, with 4/5 of the regression in the auxiliary MTP loss (**+0.47**, vs. +0.05 main).
- **Single-variable attribution** (2000 steps, same seed): per-head alone is neutral throughout ($|\Delta| \leq$ 0.09; MTP slightly better, -0.03…-0.09); EdgeCubic alone reproduces the full regression (+0.18 / +0.04 / +0.47) — additive, no interaction. The damage is localized in the MTP/shared-FFN groups — exactly the groups that received the lowest knee ($l0 = 0.0025$).

- **Mechanism discrimination** (P30, §4.3): σ^* did not drift, so staleness is not the cause; the binary rule is. The lowest-knee groups have the widest transition bands (heaviest tails, Zipf ≈ -2.4): "flatten to 1 above σ^*" hands a near-decade of 30–70%-noise directions full weight. The harm accumulates slowly and is invisible at 500 steps — the warmup-vs-steady-state lesson of stale-D, re-learned at the method level. (And re-learned a third time at the grid level: the dense 500-step knee plateau reverses by 2000 steps, §5.4.)

We retain **cubic5b05 as the default** and archive EdgeCubic behind its env flag. The corrected law suggests two rescues — knees at the *upper* band edge (which for these groups is ≈b05's position: the status quo), or per-group signal-fraction shrinkage (which is t^* itself, realized from measurements rather than a model) — but any variant must re-pass the dual-duration gate; we deliberately do not promote one here.

Why the negative result strengthens the law: the binary rule fails exactly where the corrected law says it must (lowest σ^*, widest band, most sensitive block), with an effect size (+0.47 on the component loss) that dwarfs both noise floors. A law that calls its own naive misreading's failure mode, in advance and in the right location, is doing work.

9. Related work

- **Spectral shaping in Muon**: classic NS5 (Jordan et al.); Polar Express (sharp quintic); "How Much Orthogonalization Does Muon Need?" (arXiv 2606.00371) — source of the relaxed-cubic family and the non-monotonicity observation this paper explains; Muon-p (arXiv 2606.13867) — impossibility theorem, compatible with t^* (§3.3); Pion (arXiv 2605.19282) — promotion/suppression split; tail suppression ≈ a high knee (our left arm), and their RLVR observation is the low-SNR limit of our law (§7).
- **Outside the equivariant/stateless class**: NorMuon (arXiv 2510.05491; in our frame, row-wise second moments are the first step of anisotropic-noise whitening — §4.1 measured exactly that anisotropy), Aurora (arXiv 2606.27715), RowFloor, per-head reshaping (equivariance breaking); SOAP/Shampoo, Dion's error feedback (state); Mousse (arXiv 2603.09697; curvature). The σ^* measurement is orthogonal to and composable with all of these.
- **Heavy-tailed training spectra**: HT-SR-style heavy tails are documented for weights/Hessians; we find momentum spectra are smooth heavy-tailed too — precisely what breaks MP calibration and widens the transition band.
- **RMT toolbox**: Marchenko–Pastur (1967); BBP; Benaych-Georges & Nadakuditi (2012) overlap formulas (validated synthetically at 2%); Gavish & Donoho (2017) optimal shrinkage (different objective — §3.3); von Neumann (1937) representation behind §3.1.
- **Optimizer-benchmark evidence**: the nanoGPT optimizer track's decomposable gain stack (Muon → Muon^2 → NorMuonH → u/w-floor → radial brake → RowFloor + cautious WD; 3600 → 2690 steps) corroborates that optimizer progress composes from mechanism-level fixes; this paper contributes a measurable law for one core mechanism in that stack.

10. Limitations and honest boundaries

- **Scale and window.** All probes ≤2000 micro-steps, 132M/1080M models, one codebase, single node, bf16. Stationarity of σ^* is established for micro-steps 300–1900 only; behavior at 100k+ steps, other β, and much larger batches is unknown. The $\sqrt{B}$ law failing at B/2 does not bound B/8.
- **Frames.** The gate-fix of P19 moved all absolute losses (~0.04 nat); the full five-point MoE U and the five-point dense grid both live entirely in the post-fix frame (§5.2, §5.4). Pre-fix points survive only as qualitative extremal markers (†) and cross-frame Δs are never taken.
- **Seeds.** Peak and both arms have 3 seeds each; per-seed arm penalties are uniformly positive (§5.2). Interior points near the peak (cubic5) and the dense grid remain single-seed paired comparisons. The 2000-step tier (dense P40–P42, MoE P43-vs-P28a) is single-seed, one run per point; the b002 divergence is a single event — its schedule-specificity, not its exact probability, is what the paired control establishes.

- **Estimator.** The i.i.d. reading of the ρ chance floor is approximate under the measured anisotropy; a row/column-whitened variant (NorMuon-style preprocessing) is the principled next step and may sharpen σ^*.
- **Objective proxy.** The theory optimizes per-step E[descent] - variance, not full training dynamics; all verdicts here are empirical.
- **Unmeasured region.** kv_up 128×768 shows no collapse in view ($\rho > 1$ at $\sigma_{\min} = 0.024$); the law is silent there beyond "flatten". Dense-band upper edges are early-window measurements (optimizer steps 150/300); their late-window stationarity is assumed from the MoE (P30), not re-measured.
- **Open mechanism.** The 10× dense-vs-MoE shift of the band's *lower* edge is measured, not explained; the upper region (≈1e-2–3e-2, excess never quite reaching 0.9 for attention) is common to both architectures, and why the steady-state optimum lands at knee50 ≈ 0.01 on both is consistent with, not derived from, the band picture.

11. Conclusion

The whitening knee of Muon-style optimization is governed by a measurable persistence boundary σ^* and its transition band. The boundary is read off cross-microbatch correlations with no distributional assumptions, is stationary in our training window, predicts the quality optimum across two architectures (five-point knee grids on each, three seeds on the MoE arms and peak), and explains six known failure modes plus one literature observation in one stroke. Its naive binary reading fails in a way the refined law predicts — the U-curve's arms *are* the transition band — and the arms' costs are temporally asymmetric: under-whitening is paid upfront, variance injection compounds until, at 2000 steps, it can destabilize training outright. Practically: measure $\rho(\sigma)$ before touching the knee; place the half-weight point at the band's upper edge; and distrust 500-step promotion gates — warmup optima are not steady-state optima, on both the method level (§8) and the grid level (§5.4). The closed-form RMT knee failed; the measured knee stands.

References (informal; expanded in research/SPECTRAL_THEORY.md §9 and research/OPTIMIZER_RESEARCH.md §1.3)

1. Jordan et al. — Muon. 2. Bernstein & Newhouse — Old optimizer, new norm.
2. Polar Express (arXiv 2505.16932). 4. How Much Orthogonalization Does Muon Need? (arXiv 2606.00371). 5. Pion (arXiv 2605.19282). 6. NorMuon (arXiv 2510.05491). 7. Aurora (arXiv 2606.27715). 8. Mousse (arXiv 2603.09697). 9. Muon-p (arXiv 2606.13867). 10. Shampoo; SOAP; Dion (arXiv 2504.05295). 11. Marchenko & Pastur (1967). 12. Benaych-Georges & Nadakuditi (2012). 13. Gavish & Donoho (2017). 14. von Neumann (1937).

Appendix A. Protocol and reproducibility

- All probes: seed 1337, paired data order; screening 500 micro-steps, confirmation 2000 micro-steps, log every 100; one training job at a time.
- Probe → artifact map: P13 cubic5 · P14/15 frac family · P16 b05 · P17 b10 · P18 snapshots (optimizer steps 150/300) · P19 gate-fix baseline · P20 b002 · P21/P21b per-head ± compensation · P22 batch-half · P23 EdgeCubic · P24 per-head+edge · P25 seeds · P26 dense knees (a: b05, b: b10, c: b002) · P27 dense snapshots · P28 2000-step confirmation (a: baseline, b: per-head+edge) · P29 single-variable attribution (a: per-head, b: edge) · P30 late snapshots (optimizer steps 500/750/950) · P31–P33 post-fix U re-runs (classic / cubic5 / b10) · P34/P35 dense b01/b005 · P36–P39 arm seeds (b002/b10 × 1338/1339) · P40–P42 dense 2000-step tier (b002 diverged / b05 / b005) · P43 MoE b10 2000-step check (vs. P28a). Logs in `research_runs/probe_*/console.log`; analysis dumps in `probe_p18_snap`, `probe_p27_dense_snap`, `probe_p30_latesnap` (+ `mpfit_all.txt`, `sigma_upper.txt`); queue ledger `research_runs/gap_queue_status.tsv`, `research_runs/dense_2k_status.tsv`.
- Code: `trainer/muon.py` (schedules `_CUBIC5B*_COEFFS`, `_cubic_schedule`, `_EDGE_LO_*`), `trainer/snapshot.py`, `experiments/mp_fit.py` (synthetic validation included), `experiments/sigma_upper*.py` (opera-

tional band edge, §4.3), `test_muonh_zeroinit.py` (§4.4 regression). The $\rho(\sigma)$ measurement is also packaged as a standalone numpy-only tool, `kneescope` (companion repository; MIT), with synthetic-validation tests.

- Convention: snapshot steps count optimizer updates (2 micro-steps each at accumulation 2). A unit mix-up cost one aborted run during P30; the convention is now asserted in the tooling docstring.

Appendix B. Figures

All figures render from repository artifacts: `research/figures/fig{1..4}_*.{png,pdf}`. Regenerate with `uv run --no-project --with matplotlib python experiments/paper_figs.py` (parsers read the probe logs and mp_fit outputs directly).

- **Fig.1** (existence + stationarity): $\rho(\sigma)$ per shape group, log σ; curves for optimizer steps 150/300 (warmup) and 500/750/950 overlaid; shaded transition band. Data: probe_p18_snap/mp_fit_v2.txt, probe_p30_latesnap/mpfit_all.txt.
- **Fig.2** (money): left — MoE Δloss@500 vs. knee50, five post-fix points solid (single frame), pre-fix points hollow (qualitative only), shaded transition band, 3-seed Δ error bars on the two arms; right — dense 132M five-point grid at the 500-step tier with its measured band (σ^* 10× lower, upper region ≈3e-2 shared with the MoE). Data: §5.2/§5.4 tables.
- **Fig.3** (scaling triple): per-group σ^* (log bars); batch-half pair; per-head ± compensation. Data: §4.2 table, P22, P21/P21b.
- **Fig.4** (Pareto + reversal): 500-step loss vs. step/s for b05 / EdgeCubic / per-head+{b05, edge}; 2000-step Δloss (solid) and ΔMTP (dashed) trajectories vs. the b05 baseline for ph+edge / edge / per-head. Data: P28/P29 console logs; P19/P23/P24 step/s.